A Recall AUC, all mutations
B Recall AUC, de novo
C Recall AUC, gap/stop mask
D Recall AUC, de novo + gap/stop mask
E Max precision, all mutations, 1+ cases
F Max precision, de novo, 1+ cases
G Max precision, all mutations, 100+ cases
H Max precision, de novo, 100+ cases
antiGen
Where x which + PSSM, val−selected (10x)
antiGen, val−selected (10x)
Mean of z−scores
Mean of log−probs
Mean of ranks
antiGen as Tranception full FT
antiGen as EVE full FT
antiGen as ESM 2 (650M) partial FT
antiGen as ESM 2 (650M) full FT
antiGen as ESM 2 (650M) LM head FT
antiGen as ESM 2 (35M) partial FT
antiGen as ESM 2 (35M) full FT
antiGen as ESM 2 (35M) LM head FT
BA.2 DMS (LY−CoV1404 escape)
BA.2 DMS (expression)
BA.2 DMS (bind)
BA.1 DMS (expression)
BA.1 DMS (bind)
Where × which
Substitution matrix
Tranception
EVE
MSA Trans. (BA.2 eval)
MSA Transfomer
PSSM Conservation
PSSM
Codon table
BLOSUM62
ESM 2 (650M)
ESM 2 (35M)
EVEscape
1st
2nd
3rd
AUC, recall by quantile
Max PSSM precision
Model type
antiGen
antiGen Finetunes
Deep Mutational Scans
Pandemic Data in Training Window
Pre−Pandemic Data Only
Ensembles
Val−selected, averaged over 10 holdouts